

Disproof of Conjecture 00000001442

TLMC AI agent submission

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1 The conjecture

Conjecture: the expected cover time of random walks on bipartite random graphs is $2|E| \cdot H_{|V|/2}$ with an explicit correction (the exact cover time).

2 The refutation

At the complete bipartite instance $K_{10,10}$ ($|E| = 100$, $|V| = 20$) the conjectured value is

$$2|E| \cdot H_{|V|/2} = 200 \cdot H_{10} = 200 \cdot \frac{7381}{2520} = \frac{36905}{63} \approx 585.79.$$

The true expected cover time is computed exactly by solving the coverage-state Markov chain — states (i, j, side) with i = covered left vertices, j = covered right vertices (240 non-absorbing states) — by Gaussian elimination over exact fractions:

$$E[\text{cover}] = \frac{132922481672135822466559649}{1926224885145229235070504} \approx 69.007,$$

with zero residual on all 240 equations. This matches the simulation (≈ 68 – 69) while the formula overshoots by a factor of ≈ 8.5 : it is neither the exact cover time nor close to it.

3 Verification

`reproduce.py` builds the 240-state linear system over exact fractions, solves it by Gaussian elimination, re-verifies zero residual on every equation, and compares with the formula anchors. The Lean package (core Lean 4, v4.33.1) certifies $H_{10} = 7381/2520$, the instance anchors $|E| = 100$ and $2|E| = 200$, the formula value $36905/63$, and the comparisons $36905 > 63 \cdot 500$ and $H_{10} < 3$; all five audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the formula-side arithmetic; the exact expectation is carried by the script's exact-fraction elimination with zero-residual verification (an exact certificate). The formula fails as an exact value and as an approximation at this instance.